

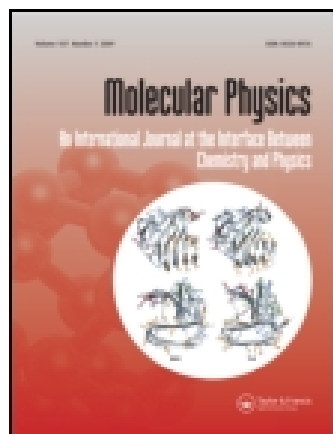
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Computer simulation and the dielectric constant at finite wavelength

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Computer simulation and the dielectric constant at finite wavelength

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In computer simulations the wavevector-dependent dielectric constant $\epsilon(k)$ may be calculated from the equilibrium fluctuations of the spatial Fourier components of the instantaneous polarization of the sample. It is shown that this relationship depends on the boundary conditions/dipolar interactions used in the simulation, not only at infinite but also at finite wavelength, and explicit expressions are given for all simulation techniques currently used. By means of these fluctuation formulas it is now possible to calculate the entire k -dependent dielectric constant from simulations with arbitrary boundary conditions. The formulas also permit a general discussion of the relative merits of various simulation techniques as well as of some quantitative aspects of the Ewald sum in particular.

Longitudinal and transverse components of $\epsilon(k)$ have been obtained from a reaction field simulation of a Stockmayer system, and these results are used to calculate the k -dependent polarization fluctuations for various other boundary conditions. The predictions are found to be consistent with the behaviour in actual simulations. Finally it is pointed out that, once the dielectric constant is known from any type of simulation, the orientational structure of the infinite system can be obtained by exploiting the relationship between $\epsilon(k)$ and the pair correlation function. It is shown that for the Stockmayer system this route is feasible in practice.

1. INTRODUCTION

In dielectric continuum theory it is usually assumed that a *local* linear relationship holds between the Maxwell field $\mathbf{E}$ and the polarization $\mathbf{P}$

$$\mathbf{P}(\mathbf{r}) = \frac{\epsilon - 1}{4\pi} \mathbf{E}(\mathbf{r}) \quad (1.1)$$

so that the value of $\mathbf{P}$ at any point within the dielectric is solely determined by the value of $\mathbf{E}$ at that single point. ϵ is the static or (if $\mathbf{E}$ and $\mathbf{P}$ vary harmonically with time) frequency dependent bulk dielectric constant, i.e. the dielectric constant at infinite wavelength. Obviously this is a very good approximation since the wavevectors $k = 2\pi/\lambda$ typical of dielectric relaxation ($k \sim 10 \text{ cm}^{-1}$) and even visible light ($k \sim 10^5 \text{ cm}^{-1}$) do indeed correspond to infinite wavelengths on a molecular scale. At shorter wavelengths, however, this relationship will not be local any

more and thus be given by a convolution [1]. In terms of spatial Fourier transforms this is written as

$$\tilde{\mathbf{P}}(\mathbf{k}) = \frac{\boldsymbol{\varepsilon}(\mathbf{k}) - I}{4\pi} \cdot \tilde{\mathbf{E}}(\mathbf{k}) \quad (1.2)$$

where the dielectric constant now is a tensor (even in an isotropic system) and depends both on the magnitude and the direction of $\mathbf{k}$.

Calculations of the k -dependent dielectric constant for simple, microscopically defined model systems would be of great theoretical interest, since this might help us understand the transition from a molecular system of statistical mechanics to a macroscopic continuum. The k -dependent dielectric constant and its associated k -dependent Kirkwood g -factors may be obtained from integral equation theories as well as from computer simulations, and some results have already become available [2]. Another reason why finite- k g -factors have sometimes been calculated in simulations [3, 4] is the fact that the zero wavevector dipole moment fluctuations, which are needed to calculate the bulk dielectric properties, suffer from large statistical error bars and are also strongly affected by the boundary conditions employed in the simulations. For fluctuations at finite wavelengths, on the other hand, the influence of the boundary conditions should be negligible, so that, by considering a few non-zero k -vectors and extrapolating to $k = 0$, one might be able to obtain a more reliable estimate of the dielectric constant.

However, as far as the boundary conditions are concerned, the problem of calculating the bulk dielectric constant in computer simulations has now been solved [2, 5–9], and today both the static and the frequency dependent dielectric constant may be reliably calculated not only for simple model systems but also for more realistic intermolecular interactions [10]. One of the main problems we have been concerned with was to show that simulations with different boundary conditions actually lead to consistent results for the dielectric constant [11]. This not only guarantees that the dielectric constant so obtained is independent of the details of the simulation, but—by definition—it is only when we are able to predict theoretically a system's macroscopic behaviour under various boundary conditions, that we may be sure to have calculated its true dielectric constant.

The purpose of the present article is to generalize our previous results to finite wavelengths and to show how the entire k -dependent (static) dielectric constant may be obtained from simulations. Using a theory introduced to account specifically for the geometries typically found in simulations, we first investigate how the boundary conditions affect the dipole moment fluctuations at finite wavelength and derive explicit expressions for all boundary conditions currently used in simulations. In these formulas the k -dependent Kirkwood g -factors are related to the k -dependent dielectric constant, and since the influence of the boundary conditions has been explicitly accounted for, they may be used to calculate the true infinite-system k -dependent dielectric constant from simulations with arbitrary boundary conditions. At the same time our results also enable us to give a more thorough characterization of the relative merits of various simulation techniques.

As a specific example, we then take the k -dependent dielectric constant obtained from a single reaction field simulation of a Stockmayer system and calculate the k -dependent g -factors that would have been observed if various

other boundary conditions had been used. As far as results from actual simulations are available, they are found to be in good agreement with our predictions.

In the final section we show how the concept of a k -dependent dielectric constant may be exploited to extract quantitatively the structure of the infinite system from a simulation using any one of the usual geometries.

2. THEORY

The following derivation is similar to that of [7] and calculates, microscopically as well as macroscopically, the response of a dipolar system to an external field. Both results are then combined to yield a microscopic expression for the dielectric constant. However, in the present case electric field and polarization are spatially inhomogeneous.

We consider an isotropic system of N identical, non-polarizable particles with dipole moment μ , enclosed at temperature T in a cubic volume V with edge length $L = V^{1/3}$, to which toroidal boundary conditions are applied. This system is placed in an external field of the form

$$\mathbf{E}_0(\mathbf{r}) = \mathbf{E}_0 \exp(i\mathbf{k} \cdot \mathbf{r}), \quad (2.1)$$

where $\mathbf{k}$ is one of the allowed wavevectors of the reciprocal lattice

$$\mathbf{k} = \frac{2\pi}{L} \mathbf{n} \quad (2.2)$$

and $\mathbf{n} = (n_x, n_y, n_z)$ is a vector with integer components. The interaction energy of the system with the external field is

$$H_1 = - \sum_{i=1}^N \boldsymbol{\mu}_i \cdot \mathbf{E}_0 \exp(i\mathbf{k} \cdot \mathbf{r}_i). \quad (2.3)$$

The spatial Fourier transform of the polarization $\mathbf{P}$ induced in the system may be written as

$$\tilde{\mathbf{P}}(\mathbf{k}) = \int_V d\mathbf{r} \exp(-i\mathbf{k} \cdot \mathbf{r}) \mathbf{P}(\mathbf{r}) = \sum_i \boldsymbol{\mu}_i \exp(-i\mathbf{k} \cdot \mathbf{r}_i) = \mathbf{M}(\mathbf{k}) \quad (2.4)$$

and is thus identical to the Fourier components $\mathbf{M}(\mathbf{k})$ of the total dipole moment which are sometimes used in the literature [3, 4].

In first-order perturbation theory the mean value of $\tilde{\mathbf{P}}$ in the presence of the external field (2.1) is given by

$$\begin{aligned} \langle \tilde{\mathbf{P}}(\mathbf{k}) \rangle_{E_0} &= \left\langle \left(\sum_i \boldsymbol{\mu}_i \exp(-i\mathbf{k} \cdot \mathbf{r}_i) \right) \left(\frac{1}{k_B T} \sum_j \boldsymbol{\mu}_j \cdot \mathbf{E}_0 \exp(i\mathbf{k} \cdot \mathbf{r}_j) \right) \right\rangle \\ &= \frac{1}{V k_B T} \left\langle \sum_{i,j} \boldsymbol{\mu}_i \boldsymbol{\mu}_j \exp(i\mathbf{k} \cdot \mathbf{r}_{ij}) \right\rangle \cdot V \mathbf{E}_0, \end{aligned} \quad (2.5)$$

where $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, and k_B is Boltzmann's constant. The last factor is just the Fourier transform $\tilde{\mathbf{E}}_0(\mathbf{k}') = V \mathbf{E}_0 \delta_{\mathbf{k}', \mathbf{k}}$, so that, using the $\mathbf{M}(\mathbf{k})$ defined above

$$\langle \tilde{\mathbf{P}}(\mathbf{k}) \rangle_{E_0} = \frac{1}{V k_B T} \langle \mathbf{M}(\mathbf{k}) \mathbf{M}(-\mathbf{k}) \rangle \cdot \tilde{\mathbf{E}}_0(\mathbf{k}). \quad (2.6)$$

If $\mathbf{E}_0$ is chosen to be either parallel or perpendicular to $\mathbf{k}$, we may write

$$\langle \tilde{\mathbf{P}}(\mathbf{k}) \rangle_{E_0} = \begin{cases} \frac{1}{V k_B T} \langle \mathbf{M}^L(\mathbf{k}) \cdot \mathbf{M}^L(-\mathbf{k}) \rangle \tilde{\mathbf{E}}_0(\mathbf{k}), & \text{if } \mathbf{E}_0 \parallel \mathbf{k} \\ \frac{1}{2V k_B T} \langle \mathbf{M}^T(\mathbf{k}) \cdot \mathbf{M}^T(-\mathbf{k}) \rangle \tilde{\mathbf{E}}_0(\mathbf{k}), & \text{if } \mathbf{E}_0 \perp \mathbf{k}. \end{cases} \quad (2.7)$$

Here $\langle \dots \rangle$ without subscript denotes an equilibrium average and $\mathbf{M}^L(\mathbf{k})$ and $\mathbf{M}^T(\mathbf{k})$ are the components parallel (longitudinal) and perpendicular (transverse) to $\mathbf{k}$. In (2.7) we have also used that—as will be shown explicitly below— $\tilde{\mathbf{P}}(\mathbf{k})$ is parallel/perpendicular to $\mathbf{k}$ whenever $\mathbf{E}_0$ has this property.

The macroscopic analogue of (2.7) may be found from the general relationship linking $\mathbf{E}_0$ and $\mathbf{P}$ to the Maxwell field $\mathbf{E}$

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\mathbf{r}) - \frac{4\pi}{3} \mathbf{P}(\mathbf{r}) + \int_V d\mathbf{r}' \mathbf{T}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}'). \quad (2.8)$$

Here, the integration extends over all V , except an infinitesimal sphere around the singularity of the dipole–dipole tensor $\mathbf{T}$ at $\mathbf{r}' = \mathbf{r}$, whose contribution is given by the second term on the right-hand side. Now, in order to describe correctly the situation typically found in simulations, equation (2.8) has to be modified in two ways [7]: The toroidal boundary conditions have to be explicitly accounted for in the integral, and more general interactions than those propagated by the usual dipole–dipole tensor

$$T_{ab}(r) = \nabla_a \nabla_b (1/r) \quad (2.9)$$

have to be allowed for. In the following these will be called modified interactions and symbolized by their T -tensor $\mathbf{T}_{\text{mod}}$ (for specific examples see [7–9] and §3).

Using (1.2) it is possible to eliminate either $\mathbf{P}$ or $\mathbf{E}$ from (2.8) in favour of the other function. However, at finite wavelengths the constitutive relation is in general a function of both the magnitude and the direction of $\mathbf{k}$, so that even for isotropic systems two principal components, defining a longitudinal and a transverse dielectric constant, $\varepsilon_L(k)$ and $\varepsilon_T(k)$, have to be distinguished, depending on whether $\mathbf{E}$ is parallel or perpendicular to $\mathbf{k}$:

$$\tilde{\mathbf{P}}(\mathbf{k}) = \begin{cases} \frac{\varepsilon_L(k) - 1}{4\pi} \tilde{\mathbf{E}}(\mathbf{k}), & \text{if } \tilde{\mathbf{E}}(\mathbf{k}) \parallel \mathbf{k}, \\ \frac{\varepsilon_T(k) - 1}{4\pi} \tilde{\mathbf{E}}(\mathbf{k}), & \text{if } \tilde{\mathbf{E}}(\mathbf{k}) \perp \mathbf{k}. \end{cases} \quad (2.10)$$

Since an arbitrary $\tilde{\mathbf{E}}(\mathbf{k})$ can always be split up into components parallel and perpendicular to $\mathbf{k}$, i.e. $\tilde{\mathbf{E}}(\mathbf{k}) = \hat{\mathbf{k}}(\hat{\mathbf{k}} \cdot \tilde{\mathbf{E}}(\mathbf{k})) + (\mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}) \cdot \tilde{\mathbf{E}}(\mathbf{k})$, we may write (2.10) in tensorial form by setting

$$\boldsymbol{\varepsilon}(\mathbf{k}) = \varepsilon_T(k) \mathbf{I} + [\varepsilon_L(k) - \varepsilon_T(k)] \hat{\mathbf{k}}\hat{\mathbf{k}}. \quad (2.11)$$

Here $\hat{\mathbf{k}}$ is the unit vector in the direction of $\mathbf{k}$, and ε_L and ε_T both depend only on the magnitude of $\mathbf{k}$.

Eliminating $\mathbf{P}$ from (2.8) by means of (1.2) and (2.11) now yields an integral equation for $\mathbf{E}$ which is best solved in Fourier space since, in addition to trans-

forming the convolution into a product, this automatically takes care of the toroidal boundary conditions [7]. The resulting algebraic equation is

$$\left\{ \frac{\varepsilon(\mathbf{k}) + 2I}{3} - \frac{\varepsilon(\mathbf{k}) - I}{3} \frac{3}{4\pi} \tilde{T}_{\text{mod}}(\mathbf{k}) \right\} \cdot \tilde{\mathbf{E}}(\mathbf{k}) = \tilde{\mathbf{E}}_0(\mathbf{k}) \quad (2.12)$$

and involves the spatial Fourier transform (2.14) of the modified T -tensor T_{mod} . In order to obtain an explicit solution for $\tilde{\mathbf{E}}(\mathbf{k})$ (and $\tilde{\mathbf{P}}(\mathbf{k})$) the inverse of the expression in curly brackets has to be found.

At this point we have to make an assumption about the general form of T_{mod} . From the examples discussed in §3 it will become clear that an interaction of the type

$$T_{\text{mod}}(\mathbf{r}) = a(r)(3\hat{\mathbf{r}}\hat{\mathbf{r}} - I) + b(r)I \quad (2.13)$$

is sufficiently general to cover all simulation techniques currently used. In (2.13) the angular dependence of T_{mod} is represented by the tensors $(3\hat{\mathbf{r}}\hat{\mathbf{r}} - I)$ and I . $\hat{\mathbf{r}}$ is the unit vector in the direction of $\mathbf{r}$, and $a(r)$ and $b(r)$ are functions depending only on the magnitude of $\mathbf{r}$. The usual dipole-dipole interaction (2.9) is recovered by setting $a(r) = 1/r^3$ for all $0 < r < \infty$ and $b(r) \equiv 0$. In general $b(r)$ serves to represent a reaction field.

Expressing $(3\hat{\mathbf{r}}\hat{\mathbf{r}} - I)$ and I as products of spherical harmonics of Ω_r , the angles specifying the orientation of $\mathbf{r}$, and inserting the Rayleigh expansion for $\exp(-i\mathbf{k} \cdot \mathbf{r})$, one finds that the Fourier transform of (2.13)

$$\tilde{T}_{\text{mod}}(\mathbf{k}) = \int_0^\infty dr r^2 \int d\Omega_r \exp(-i\mathbf{k} \cdot \mathbf{r}) T_{\text{mod}}(\mathbf{r}) \quad (2.14)$$

may be written as

$$\tilde{T}_{\text{mod}}(\mathbf{k}) = \frac{4\pi}{3} [A(k)(3\hat{\mathbf{k}}\hat{\mathbf{k}} - I) + B(k)I], \quad (2.15)$$

with $A(k)$ and $B(k)$ given by

$$A(k) = -3 \int_0^\infty dr r^2 j_2(kr) a(r), \quad (2.16 a)$$

$$B(k) = 3 \int_0^\infty dr r^2 j_0(kr) b(r). \quad (2.16 b)$$

Here $j_0(x)$ and $j_2(x)$ are the spherical Bessel functions of order 0 and 2, respectively, and the singularity of T_{mod} need not be considered since it has been explicitly accounted for in (2.8). Thus (2.15) is of the same general form, in Fourier space, as (2.13). The angular dependence is contained in $(3\hat{\mathbf{k}}\hat{\mathbf{k}} - I)$ and I , and $A(k)$ and $B(k)$ are functions of the magnitude of $\mathbf{k}$ only.

Alternatively, $\tilde{T}_{\text{mod}}(\mathbf{k})$ may be written as a linear combination of $\hat{\mathbf{k}}\hat{\mathbf{k}}$ and I , which is how we have chosen to represent the dielectric tensor $\varepsilon(\mathbf{k})$, equation (2.11). The expression in curly brackets in (2.12) is now seen to be made up of sums and products of $\hat{\mathbf{k}}\hat{\mathbf{k}}$ and I , and since products of $\hat{\mathbf{k}}\hat{\mathbf{k}}$ and I yield either $\hat{\mathbf{k}}\hat{\mathbf{k}}$ or I again, it will also be of the general form

$$\{ \dots \} = XI + Y\hat{\mathbf{k}}\hat{\mathbf{k}} \quad (2.17)$$

with coefficients X and Y depending only on the magnitude of $\mathbf{k}$. Of course, the inverse is also a linear combination of $\hat{\mathbf{k}}\hat{\mathbf{k}}$ and $\mathbf{I}$, and one finds

$$\{\dots\}^{-1} = \frac{1}{X} \mathbf{I} - \frac{Y}{X(X+Y)} \hat{\mathbf{k}}\hat{\mathbf{k}}. \quad (2.18)$$

We now consider the two particular situations in which the applied field is either purely longitudinal or purely transverse. If $\mathbf{E}_0$ is purely longitudinal, i.e. $\tilde{\mathbf{E}}_0(\mathbf{k}) \equiv \hat{\mathbf{k}}(\hat{\mathbf{k}} \cdot \tilde{\mathbf{E}}_0(\mathbf{k}))$, solving (2.12) by means of (2.18) yields

$$\tilde{\mathbf{E}}(\mathbf{k}) = \{\dots\}^{-1} \cdot \hat{\mathbf{k}}\hat{\mathbf{k}} \cdot \tilde{\mathbf{E}}_0(\mathbf{k}) = \frac{1}{X+Y} \hat{\mathbf{k}}\hat{\mathbf{k}} \cdot \tilde{\mathbf{E}}_0(\mathbf{k}) \equiv \frac{1}{X+Y} \tilde{\mathbf{E}}_0(\mathbf{k}), \quad (2.19 a)$$

while for $\mathbf{E}_0$ transverse, i.e. $\tilde{\mathbf{E}}_0(\mathbf{k}) \equiv (\mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}) \cdot \tilde{\mathbf{E}}_0(\mathbf{k})$, we obtain

$$\tilde{\mathbf{E}}(\mathbf{k}) = \{\dots\}^{-1} \cdot (\mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}) \cdot \tilde{\mathbf{E}}_0(\mathbf{k}) = \frac{1}{X} (\mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}) \cdot \tilde{\mathbf{E}}_0(\mathbf{k}) \equiv \frac{1}{X} \tilde{\mathbf{E}}_0(\mathbf{k}). \quad (2.19 b)$$

In this last step two things have been achieved: On the one hand, we have shown that—as claimed above— $\mathbf{E}$ and (by virtue of (2.10)) $\mathbf{P}$ are both longitudinal whenever $\mathbf{E}_0$ is longitudinal, and transverse when $\mathbf{E}_0$ is transverse. On the other hand, we have also found the explicit solution of (2.12). In fact, inserting (2.11) and (2.15) into (2.12) and evaluating the coefficients X and Y , we obtain from (2.19), switching also from $\mathbf{E}$ to $\mathbf{P}$ by means of (2.10),

$$\tilde{\mathbf{P}}(\mathbf{k}) = \frac{3}{4\pi} \frac{\varepsilon_L(k) - 1}{\varepsilon_L(k) + 2} \left\{ 1 - \frac{\varepsilon_L(k) - 1}{\varepsilon_L(k) + 2} [2A(k) + B(k)] \right\}^{-1} \tilde{\mathbf{E}}_0(\mathbf{k}), \quad \text{if } \mathbf{E}_0 \parallel \mathbf{k}, \quad (2.20 a)$$

$$\tilde{\mathbf{P}}(\mathbf{k}) = \frac{3}{4\pi} \frac{\varepsilon_T(k) - 1}{\varepsilon_T(k) + 2} \left\{ 1 - \frac{\varepsilon_T(k) - 1}{\varepsilon_T(k) + 2} [B(k) - A(k)] \right\}^{-1} \tilde{\mathbf{E}}_0(\mathbf{k}), \quad \text{if } \mathbf{E}_0 \perp \mathbf{k}. \quad (2.20 b)$$

This is the polarization induced by $\mathbf{E}_0$ in the sample, as calculated by macroscopic electrostatics.

Combining the macroscopic result with its statistical-mechanical counterpart (2.7) we finally obtain

$$\frac{\varepsilon_L(k) - 1}{\varepsilon_L(k) + 2} \left\{ 1 - \frac{\varepsilon_L(k) - 1}{\varepsilon_L(k) + 2} [2A(k) + B(k)] \right\}^{-1} = 3yG^L(k), \quad (2.21 a)$$

$$\frac{\varepsilon_T(k) - 1}{\varepsilon_T(k) + 2} \left\{ 1 - \frac{\varepsilon_T(k) - 1}{\varepsilon_T(k) + 2} [B(k) - A(k)] \right\}^{-1} = 3yG^T(k), \quad (2.21 b)$$

where

$$G^L(k) = \langle \mathbf{M}^L(\mathbf{k}) \cdot \mathbf{M}^L(-\mathbf{k}) \rangle / N\mu^2, \quad (2.22 a)$$

$$G^T(k) = \langle \mathbf{M}^T(\mathbf{k}) \cdot \mathbf{M}^T(-\mathbf{k}) \rangle / 2N\mu^2 \quad (2.22 b)$$

and ($\rho = N/V$)

$$y = \frac{4\pi\rho\mu^2}{9k_B T} \quad (2.23)$$

$G^L(k)$ and $G^T(k)$ are, respectively, the longitudinal and transverse components of

$$G(k) = G^L(k) + 2G^T(k) = \langle \mathbf{M}(\mathbf{k}) \cdot \mathbf{M}(-\mathbf{k}) \rangle / N\mu^2. \quad (2.24)$$

This is the k -dependent generalization of the finite-system g -factor

$$G_K = \langle M^2 \rangle / N\mu^2 \quad (2.25)$$

to which it reduces as $k \rightarrow 0$

$$\lim_{k \rightarrow 0} G(k) = \lim_{k \rightarrow 0} [G^L(k) + 2G^T(k)] = \langle \mathbf{M}(0) \cdot \mathbf{M}(0) \rangle / N\mu^2 = G_K. \quad (2.26)$$

(Note that we are using upper case letters for the properties of a finite system while lower case letters are reserved for the infinite system, e.g. ' g_K ' for the Kirkwood g -factor in its usual sense [12].)

Equation (2.21) is our central result. In it the k -dependent dielectric constants $\varepsilon_L(k)$ and $\varepsilon_T(k)$ are related to the longitudinal and transverse components of the finite-system g -factor, i.e. to the finite wavelength equilibrium dipole moment fluctuations of the system. That a relationship such as (2.21) exists is not new, of course, and similar expressions can be found in [2] and [4]. What is new, however, is the fact that our derivation is specifically adapted to the geometries typical of computer simulations of polar systems. In the present work both the toroidal boundary conditions and the influence of the particular form of dipolar interactions have been taken into account, the former by formulating, and solving, the fundamental equation (2.8) of electrostatics in k -space, so that they are implicitly contained in our result. The influence of the latter, i.e. of the modified T -tensor used in the simulation, can be treated explicitly and gives rise in (2.21) to the terms containing $A(k)$ and $B(k)$. Thus, if we know $A(k)$ and $B(k)$ for a given simulation technique, then equation (2.21) may be used to calculate the entire k -dependent dielectric constant.

When interpreted in a different way, i.e. assuming that $\varepsilon(\mathbf{k})$ is known and asking for the effect of various forms of $\mathbf{T}_{\text{mod}}$, equation (2.21) also tells us that in a finite system dipole moment fluctuations in general will depend upon boundary conditions not only at $k = 0$ but also at finite wavelengths. Although, for non-zero k , this effect would be expected to become negligible if the system could be made sufficiently large, it has been quantitatively accounted for; and the k -dependent dielectric constant calculated by means of (2.21) should be independent of the details of the simulation, i.e. the true dielectric constant of an infinite system.

3. RESULTS FOR SPECIFIC BOUNDARY CONDITIONS

In order to be applicable to actual simulations, equation (2.21) still needs to be complemented by explicit expressions for the functions $A(k)$ and $B(k)$ characterizing the Fourier transform of the dipole interaction tensor $\mathbf{T}_{\text{mod}}$. In the present section we will calculate $A(k)$ and $B(k)$ for simulations in reaction field (RF) geometry as well as for simulations using lattice sums (LS), such as the Ewald sum (EK), and these simulation techniques will be discussed in some detail. For the purpose of later reference, however, we first present the results for an infinite system.

3.1. The infinite system

The k -dependent g -factors for an infinite system will be denoted by $g^L(k)$, $g^T(k)$, and $g(k)$, and may be obtained by formally letting $N, V \rightarrow \infty$ in (2.22). This automatically yields the correct results for $k \neq 0$, but the results will also be correct for $k = 0$ if we agree to always let $V \rightarrow \infty$ before taking the limit $k \rightarrow 0$. In this way surface effects are eliminated and the correlations obtained are those typical of an infinite system.

In an infinite system dipoles interact through the usual (unmodified) T -tensor (2.9) characterized by

$$a(r) = 1/r^3, \quad b(r) \equiv 0. \quad (3.1)$$

Inserting this into (2.16) and integrating from 0 to ∞ (since $V \rightarrow \infty$) we obtain

$$A(k) \equiv -1, \quad B(k) \equiv 0. \quad (3.2)$$

For this pair of functions (2.21) now yields the well-known result [2]

$$\frac{\varepsilon_L(k) - 1}{3\varepsilon_L(k)} = 3yg^L(k), \quad (3.3 a)$$

$$\frac{\varepsilon_T(k) - 1}{3} = 3yg^T(k), \quad (3.3 b)$$

which may also be written as

$$1/\varepsilon_L(k) = 1 - 9yg^L(k), \quad (3.4 a)$$

$$\varepsilon_T(k) = 1 + 9yg^T(k). \quad (3.4 b)$$

At this point a number of simple results may be derived: Since g -factors by construction cannot be negative (cf. (2.4) and (2.22)), the last pair of equations implies that in general

$$\varepsilon_L(k) \geq 1 \quad \text{or} \quad \varepsilon_L(k) \leq 0, \quad (3.5 a)$$

$$\varepsilon_T(k) \geq 1. \quad (3.5 b)$$

Moreover, since there can be no correlation between different particles i and j (cf. (2.5)) at very short wavelengths, the right-hand side of (2.22) in this limit will be given by the sum of the self-terms, i.e.

$$\lim_{k \rightarrow \infty} g^L(k) = \lim_{k \rightarrow \infty} g^T(k) = 1/3, \quad (3.6)$$

which leads to the Onsager limits

$$\lim_{k \rightarrow \infty} \varepsilon_L(k) = 1/(1 - 3y), \quad (3.7 a)$$

$$\lim_{k \rightarrow \infty} \varepsilon_T(k) = 1 + 3y. \quad (3.7 b)$$

We also have

$$\lim_{k \rightarrow \infty} \{1/\varepsilon_L(k) + \varepsilon_T(k)\} = 2, \quad (3.8)$$

which indicates again that $\varepsilon_L(k)$ may take on negative values.

In the opposite limit, i.e. for $k \rightarrow 0$, $g^L(k)$ and $g^T(k)$ are expected to behave quite differently. In fact, assuming that $\varepsilon_L(k)$ and $\varepsilon_T(k)$ are both continuous at $k = 0$ and tend to the bulk dielectric constant ε_0 , equation (3.3) implies

$$g^L(0) = \frac{\varepsilon_0 - 1}{9\varepsilon_0 y} = \frac{1}{2\varepsilon_0 + 1} g_K, \quad (3.9 a)$$

$$g^T(0) = \frac{\varepsilon_0 - 1}{9y} = \frac{\varepsilon_0}{2\varepsilon_0 + 1} g_K. \quad (3.9 b)$$

Thus the transverse g -factor is proportional to the bulk dielectric constant, whereas the longitudinal component is bounded and usually rather small. g_K is the usual Kirkwood g -factor [12] and may be written as the $k = 0$ limit of $g(k) = g^L(k) + 2g^T(k)$

$$g_K \equiv g(0) = g^L(0) + 2g^T(0) = \frac{(2\varepsilon_0 + 1)(\varepsilon_0 - 1)}{9\varepsilon_0 y}. \quad (3.10)$$

In practice the g -factors for an infinite system are not, of course, obtained from actual dipole moment fluctuations, but from the components $h_\Delta(r)$ and $h_D(r)$ [1, 2] of the angle-dependent pair correlation function. The relevant formulas are

$$g^L(k) = \frac{1}{3} + \frac{1}{3}[\rho\tilde{h}_\Delta(k) - 2\rho\tilde{h}_D(k)], \quad (3.11 a)$$

$$g^T(k) = \frac{1}{3} + \frac{1}{3}[\rho\tilde{h}_\Delta(k) + \rho\tilde{h}_D(k)], \quad (3.11 b)$$

where

$$\rho\tilde{h}_\Delta(k) = 4\pi\rho \int_0^\infty dr r^2 j_0(kr) h_\Delta(r), \quad (3.12 a)$$

$$\rho\tilde{h}_D(k) = 4\pi\rho \int_0^\infty dr r^2 j_2(kr) h_D(r). \quad (3.12 b)$$

(A derivation of (3.11–12) is given in the Appendix.) From (3.11) we note that $g(k)$ is essentially the Fourier transform of $h_\Delta(r)$

$$g(k) = g^L(k) + 2g^T(k) = 1 + \frac{\rho}{3} \tilde{h}_\Delta(k) \quad (3.13)$$

and in particular

$$g_K = 1 + \frac{\rho}{3} \tilde{h}_\Delta(0). \quad (3.14)$$

Having discussed the infinite system we now consider finite toroidal systems and present specific results that are useful in the analysis of computer simulations of polar systems. Today these simulations are usually carried out using either the reaction field method or the Ewald (lattice) sum.

3.2. Reaction field geometry

In simulations using the reaction field method the dipole–dipole interaction is truncated at some cutoff radius $r_c \leq L/2$; the long-range interactions that are

neglected in this way are approximated by Onsager's reaction field [13]. This may be interpreted as a modified interaction, characterized by a T -tensor

$$\mathbf{T}_{\text{RF}}(\mathbf{r}) = \begin{cases} \mathbf{T}(\mathbf{r}) + \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{1}{r_c^3} \mathbf{I}, & \text{if } r < r_c, \\ 0, & \text{otherwise,} \end{cases} \quad (3.15)$$

where ϵ_{RF} is the dielectric constant of the background continuum. Equation (3.15) is of the general form of (2.13), with the radial functions given by

$$\begin{aligned} a(r) &= 1/r^3, \quad b(r) = \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{1}{r_c^3}, & \text{if } r < r_c, \\ a(r) &= b(r) = 0, & \text{otherwise.} \end{aligned} \quad (3.16)$$

Inserting these into (2.16) and noting that the integrals are only over a finite range, we obtain

$$A(k) = -1 + \frac{3j_1(kr_c)}{kr_c}, \quad (3.17a)$$

$$B(k) = \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{3j_1(kr_c)}{kr_c}. \quad (3.17b)$$

(Here $j_1(x) = \sin x/x^2 - \cos x/x$ is the spherical Bessel function of order one with $\lim_{x \rightarrow 0} j_1(x)/x = 1/3$ and $\lim_{x \rightarrow \infty} j_1(x) = 0$. The results for the infinite system, equation (3.2), are recovered from (3.17) by letting $r_c \rightarrow \infty$.)

For reaction field geometry this represents already the most general result. Having been complemented by (3.17), equation (2.21) may now be used to calculate the k -dependent dielectric constants $\epsilon_L(k)$ and $\epsilon_T(k)$ from the finite-system g -factors $G^L(k)$ and $G^T(k)$ obtained in the simulation. From the dielectric constant we may in turn calculate the g -factors of the infinite system, since they are related to it through (3.3). In general there exists no simple *direct* relationship between the g -factors of the finite system and those of the infinite system, except for the following special values of k and r_c .

For r_c fixed and $k \rightarrow 0$ we have $A(k) \rightarrow 0$, so that in this limit $2A(k) + B(k)$ and $B(k) - A(k)$ take on the same value $B(0) = 2(\epsilon_{\text{RF}} - 1)/(2\epsilon_{\text{RF}} + 1)$. Assuming again that $\epsilon_L(k)$ and $\epsilon_T(k)$ both tend to the isotropic bulk dielectric constant ϵ_0 as $k \rightarrow 0$, we find that in reaction field simulations $G^L(0)$ and $G^T(0)^\dagger$ are always identical, and equal to one third of the finite-system g -factor (2.25)

$$\begin{aligned} G^L(0) &= G^T(0) = \frac{2\epsilon_{\text{RF}} + 1}{2\epsilon_{\text{RF}} + \epsilon_0} \frac{\epsilon_0 - 1}{9y}, \\ &= \frac{1}{3} G_K = \frac{2\epsilon_{\text{RF}} + 1}{2\epsilon_{\text{RF}} + \epsilon_0} \frac{\epsilon_0}{2\epsilon_0 + 1} g_K, \end{aligned} \quad (3.18)$$

[†] Note that a decomposition of $\mathbf{M}(\mathbf{k})$ into longitudinal and transverse components cannot be effected for $k = 0$. We therefore *define* $G^L(0)$ and $G^T(0)$ by the value of (2.21) when $A(0)$, $B(0)$, and $\epsilon_L(0) = \epsilon_T(0) = \epsilon_0$ are inserted on the left-hand side. In this way $G^L(k)$ and $G^T(k)$ will be continuous/discontinuous at $k = 0$ whenever $A(k)$ and/or $B(k)$ has this property (cf. § 3.3). By contrast, $G(0)$ is always defined by (2.26).

(This last result is in agreement with the direct calculation of $G_{\mathbf{k}}$, equation (25) in [9].) Thus the behaviour of the finite-system g -factors is quite different from that of $g^L(k)$ and $g^T(k)$, which were found to be not equal at $k = 0$, with $g^L(0)$ in general being much smaller than $g^T(0)$.

We now consider special values of the parameter ε_{RF} . By setting $\varepsilon_{\text{RF}} = 1$ the reaction field in (3.15) is turned off, and interactions are completely ignored for $r > r_c$. This is known as spherical cutoff (SC) geometry and is the standard method in simulations of non-polar systems. Since $B(k) \equiv 0$ in this case, simulations in SC geometry may be analysed by using just the $A(k)$ of equation (3.17 *a*), but this will only work for very weakly polar systems, for two reasons: On the one hand, $G^L(k)$ and $G^T(k)$ are found to obey a Clausius–Mossotti-like equation near $k = 0$

$$G^L(0) = G^T(0) = \frac{1}{3y} \frac{\varepsilon_0 - 1}{\varepsilon_0 + 2}, \quad (3.19)$$

which, for statistical reasons, is difficult to solve for ε_0 if ε_0 is large. On the other hand, truncation of the dipole–dipole interaction, together with the toroidal boundary conditions, so severely distorts the structure of even moderately polar systems that this method is generally not recommended.

Choosing $\varepsilon_{\text{RF}} = \varepsilon_0$ would appear to correspond most closely to the original intention of the reaction field method, i.e. surrounding the sample with a continuum of its own dielectric constant. Since (3.18) in this case reduces to

$$G^L(0) = G^T(0) = \frac{1}{3} G_{\mathbf{k}} = \frac{(2\varepsilon_0 + 1)(\varepsilon_0 - 1)}{27\varepsilon_0 y} = \frac{1}{3} g_{\mathbf{k}} \quad (3.20)$$

this method does indeed reproduce the (Kirkwood) g -factor of the infinite system, i.e. $G_{\mathbf{k}} = g_{\mathbf{k}}$, but, individually, its longitudinal and transverse components $G^L(0)$ and $G^T(0)$ are not identical to $g^L(0)$ and $g^T(0)$ as given by (3.9). This is also true for $k \neq 0$, where $G^L(k)$, $G^T(k)$, and even $G(k)$ in general are different from the corresponding g -factors of the infinite system. Nevertheless, the latter, as well as the dielectric constant itself, may easily be calculated from the simulation by using (2.21) together with (3.17) (with the proper ε_{RF}) and then (3.3).

Another particularly interesting case is obtained in the limit $\varepsilon_{\text{RF}} \rightarrow \infty$ (conducting boundary conditions). In this case we have from (3.17) that $B(k) - A(k) \equiv 1$, independent of k , and (2.21 *b*) simply reads

$$\varepsilon_T(k) = 1 + 9yG^T(k) \quad (3.21)$$

(cf. (3.4 *b*)). Since this is just the result for the infinite system, the transverse g -factor obtained from RF simulations with conducting boundary conditions is identical, for all $k \geq 0$, to the transverse g -factor of the infinite system, i.e. $G^T(k) = g^T(k)$. It may therefore be taken directly from the simulation. $\varepsilon_L(k)$ and $g^L(k)$, on the other hand, still have to be calculated (indirectly) from $G^L(k)$ by means of the general relations (3.17 *a*), (2.21 *a*), and (3.3 *a*). At $k = 0$, in particular, we have

$$G^L(0) = G^T(0) = \frac{1}{3} G_{\mathbf{k}} = \frac{\varepsilon_0 - 1}{9y} = \frac{\varepsilon_0}{2\varepsilon_0 + 1} g_{\mathbf{k}} \quad (3.22)$$

so that $G^T(0) = g^T(0)$ as expected, but $G^L(0)$ and G_K are both different from the infinite-system g -factors $g^L(0)$ and g_K .

3.3. Lattice sums

In simulations using lattice summation techniques each particle in the basic cell interacts not only with its neighbours in the basic cell, but also with all their images in an infinite array of identical replicas of the basic cell. This may be interpreted as an effective interaction involving just the particles in the basic cell, characterized by a modified T -tensor

$$T_{LS}(\mathbf{r}) = \sum_{\mathbf{n}} T(\mathbf{r} + \mathbf{n}L) + \frac{4\pi}{3V} \frac{2(\varepsilon_{RF} - 1)}{2\varepsilon_{RF} + 1} \mathbf{I}. \quad (3.23)$$

Here, the sum is over all integer vectors $\mathbf{n}$, and the second term represents an explicit reaction field which is sometimes added to the lattice sum. Due to the long-range nature of the dipole-dipole interaction (2.9) the sum in (3.23) is only conditionally convergent, and its value depends on the order of summation [5]. In the following we will therefore assume that the summation is over spherical shells $|\mathbf{n}| \leq \mathcal{N}$, i.e. the basic cell is imagined to be at the centre of a finite spherical lattice K of identical replicas, and we then let the radius of K go to infinity.

In order to calculate the Fourier transform of (3.23) we proceed as in [8]. Starting with the first term of (3.23) we note that, since $\exp(-i\mathbf{k} \cdot \mathbf{r})$ is periodic, the integral over V of the *modified* T -tensor, is identical to the integral of the *original* T -tensor, but taken over K

$$\int_V d\mathbf{r} \exp(-i\mathbf{k} \cdot \mathbf{r}) \sum_{|\mathbf{n}| \leq \mathcal{N}} T(\mathbf{r} + \mathbf{n}L) = \int_K d\mathbf{r} \exp(-i\mathbf{k} \cdot \mathbf{r}) T(\mathbf{r}). \quad (3.24)$$

In the limit $\mathcal{N} \rightarrow \infty$, i.e. $K \rightarrow R^3$, the right-hand side of (3.24) is the Fourier transform of $T(\mathbf{r})$ over an infinite sphere, and we obtain $A(k) \equiv -1$ as in §3.1, but only for $k \neq 0$. For $k = 0$, i.e. $\exp(-i\mathbf{k} \cdot \mathbf{r}) = 1$, the integral has to be calculated separately and, taking advantage of the spherical symmetry of K as well as of the angle-dependence of T , is easily found to vanish. Thus $A(k)$ is discontinuous at $k = 0$ and given by

$$A(k) = \begin{cases} 0, & \text{for } k = 0, \\ -1, & \text{for } k \neq 0. \end{cases} \quad (3.25)$$

The second term in (3.23), representing a cubic reaction field, is constant within V so that its Fourier transform

$$B(k) = \begin{cases} \frac{2(\varepsilon_{RF} - 1)}{2\varepsilon_{RF} + 1}, & \text{for } k = 0, \\ 0, & \text{for } k \neq 0 \end{cases} \quad (3.26)$$

is concentrated at $k = 0$. (In RF geometry a more complicated $B(k)$ results, since there the reaction field is truncated spherically.) Thus the linear combinations needed in (2.21) are given by

$$\left. \begin{aligned} 2A(k) + B(k) \\ B(k) - A(k) \end{aligned} \right\} = \frac{2(\varepsilon_{\text{RF}} - 1)}{2\varepsilon_{\text{RF}} + 1}, \quad \text{for } k = 0, \quad (3.27 a)$$

$$\begin{aligned} 2A(k) + B(k) &= -2, \\ B(k) - A(k) &= 1, \end{aligned} \quad \text{for } k \neq 0. \quad (3.27 b)$$

We note that $B(k) - A(k)$, i.e. $G^T(k)$, can be made continuous by letting $\varepsilon_{\text{RF}} \rightarrow \infty$, as is usually done in practice. On the other hand, since $2A(k) + B(k)$ remains discontinuous for *any* $1 \leq \varepsilon_{\text{RF}} \leq \infty$, $G^L(k)$ will always be discontinuous at $k = 0$.

Inserting (3.27 b) into (2.21) we now obtain, for $k \neq 0$,

$$1/\varepsilon_L(k) = 1 - 9yG^L(k) \quad (3.28 a)$$

$$\varepsilon_T(k) = 1 + 9yG^T(k) \quad (3.28 b)$$

which looks just like the result for the infinite system, equations (3.4). Thus for $k \neq 0$ the finite-system g -factors obtained from simulations using the lattice sum (3.23) are already identical to the g -factors for the infinite system, i.e. $G^L(k) = g^L(k)$ and $G^T(k) = g^T(k)$. This result is independent of the value of ε_{RF} .

For $k = 0$ the situation is different, however: From (3.27 a) it is clear that also for lattice sums longitudinal and transverse g -factors are equal at $k = 0$, i.e. $G^L(0) = G^T(0) = G_K/3$ as in RF geometry (cf. §3.2). In fact, even the values are identical to those predicted by (3.18) for (spherical) RF geometry with the same ε_{RF} . By choosing suitable values for ε_{RF} we may thus achieve that either the total g -factor is identical to the infinite-system (Kirkwood) g -factor, i.e. $G_K = g_K$ (for $\varepsilon_{\text{RF}} = \varepsilon_0$); or that the transverse component is equal to the infinite-system value, i.e. $G^T(0) = g^T(0)$ (for $\varepsilon_{\text{RF}} = \infty$). In the latter case $G^T(k)$ would then be continuous at $k = 0$ and therefore completely identical to $g^T(k)$ for *all* $k \geq 0$. By contrast, there exists no (reasonable) value of ε_{RF} for which $G^L(0) = g^L(0)$, so that $G^L(k)$ will always be discontinuous at $k = 0$.

3.4. The Ewald sum

Strictly speaking, the results obtained above apply only to the idealized sum (3.23). In practice, however, most simulations use a representation going back to Ewald and Kornfeld [14], which corresponds to the case $\varepsilon_{\text{RF}} = \infty$ and replaces (3.23) by two infinite sums: one in r -space, and one in k -space, whose relative weights are controlled by a parameter η . The total sum is independent of η and identical to (3.23), but values of η can be found for which both sums converge very quickly and may be truncated after a finite number of terms.

In order to calculate how these approximations affect the results of §3.3, we write the modified T -tensor for the Ewald method as a sum of three terms

$$T_{\text{EK}} = T_{\text{EK}}^{\text{I}} + T_{\text{EK}}^{\text{II}} + T_{\text{EK}}^{\text{III}}, \quad (3.29)$$

where

$$\mathbf{T}_{\text{EK}}^{\text{I}}(\mathbf{r}) = -\frac{4\pi}{V} \sum_{0 \leq |\mathbf{k}| \leq k_{\text{max}}} \exp(-k^2/4\eta^2) \exp(-i\mathbf{k} \cdot \mathbf{r}) \hat{\mathbf{k}}\hat{\mathbf{k}}, \quad (3.30 a)$$

$$\mathbf{T}_{\text{EK}}^{\text{II}}(\mathbf{r}) = \left\{ \text{erfc}(\eta r) + \left[\frac{2\eta r}{\sqrt{\pi}} + \frac{4\eta^3 r^3}{3\sqrt{\pi}} \right] \exp(-\eta^2 r^2) \right\} \mathbf{T}(\mathbf{r}), \quad (3.30 b)$$

$$\mathbf{T}_{\text{EK}}^{\text{III}}(\mathbf{r}) = \frac{4\eta^3}{3\sqrt{\pi}} \exp(-\eta^2 r^2) \mathbf{I}. \quad (3.30 c)$$

In (3.30) it has been assumed that real space sum and Fourier space sum are both truncated spherically. Thus the sum in k -space, $\mathbf{T}_{\text{EK}}^{\text{I}}$, is only over k -vectors (2.2) with $0 \leq |\mathbf{k}| \leq k_{\text{max}}$. In the case of the real space sum we have additionally assumed that it may be truncated within the basic cell, so that $\mathbf{T}_{\text{EK}}^{\text{II}}$ and $\mathbf{T}_{\text{EK}}^{\text{III}}$ both reduce to a single term each. Hence we also have $\mathbf{T}_{\text{EK}}^{\text{II}}(\mathbf{r}) = \mathbf{T}_{\text{EK}}^{\text{III}}(\mathbf{r}) = 0$ for $r > r_c$, even though this has not been indicated explicitly in (3.30 b–c).

The Fourier transforms of (3.30) are

$$\tilde{\mathbf{T}}_{\text{EK}}^{\text{I}}(\mathbf{k}) = \begin{cases} 4\pi \exp(-k^2/4\eta^2) \hat{\mathbf{k}}\hat{\mathbf{k}}, & \text{if } 0 \leq |\mathbf{k}| \leq k_{\text{max}}, \\ 0, & \text{otherwise} \end{cases} \quad (3.31 a)$$

(since $\mathbf{T}_{\text{EK}}^{\text{I}}$ in (3.30 a) already is in Fourier representation). $\mathbf{T}_{\text{EK}}^{\text{II}}$ and $\mathbf{T}_{\text{EK}}^{\text{III}}$, on the other hand, are of the general form of (2.13), so reading off $a(r)$ and $b(r)$ and using (2.15) and (2.16) we obtain

$$\tilde{\mathbf{T}}_{\text{EK}}^{\text{II}}(\mathbf{k}) = -4\pi \int_0^{r_c} dr \frac{j_2(kr)}{r} \left\{ \text{erfc}(\eta r) + \left[\frac{2\eta r}{\sqrt{\pi}} + \frac{4\eta^3 r^3}{3\sqrt{\pi}} \right] \exp(-\eta^2 r^2) \right\} \times (3\hat{\mathbf{k}}\hat{\mathbf{k}} - \mathbf{I}), \quad (3.31 b)$$

$$\tilde{\mathbf{T}}_{\text{EK}}^{\text{III}}(\mathbf{k}) = \frac{16\sqrt{\pi}\eta^3}{3} \int_0^{r_c} dr r^2 j_0(kr) \exp(-\eta^2 r^2) \mathbf{I}. \quad (3.31 c)$$

By adding a term $(4\pi/3) \exp(-k^2/4\eta^2) \mathbf{I}$ to $\tilde{\mathbf{T}}_{\text{EK}}^{\text{I}}(\mathbf{k})$ and subtracting the same term from $\tilde{\mathbf{T}}_{\text{EK}}^{\text{III}}(\mathbf{k})$, it is now easy to see that $\tilde{\mathbf{T}}_{\text{EK}}(\mathbf{k})$ may be written in the form of (2.15) so that the formalism of § 2 is applicable again.

Thus the functions $2A(k) + B(k)$ and $B(k) - A(k)$ which are needed to calculate $\varepsilon_{\text{L}}(k)$ and $\varepsilon_{\text{T}}(k)$, are from (3.31) found to be

$$\begin{aligned} 2A(k) + B(k) = & -3 \exp(-k^2/4\eta^2) + \frac{4\eta^3}{\sqrt{\pi}} \int_0^{r_c} dr r^2 j_0(kr) \exp(-\eta^2 r^2) \\ & - 6 \int_0^{r_c} dr \frac{j_2(kr)}{r} \left\{ \text{erfc}(\eta r) + \left[\frac{2\eta r}{\sqrt{\pi}} + \frac{4\eta^3 r^3}{3\sqrt{\pi}} \right] \exp(-\eta^2 r^2) \right\}, \end{aligned} \quad (3.32 a)$$

where the first term on the right-hand side is only present for $0 \leq |\mathbf{k}| \leq k_{\text{max}}$, and

$$\begin{aligned} B(k) - A(k) = & \frac{4\eta^3}{\sqrt{\pi}} \int_0^{r_c} dr r^2 j_0(kr) \exp(-\eta^2 r^2) \\ & + 3 \int_0^{r_c} dr \frac{j_2(kr)}{r} \left\{ \text{erfc}(\eta r) + \left[\frac{2\eta r}{\sqrt{\pi}} + \frac{4\eta^3 r^3}{3\sqrt{\pi}} \right] \exp(-\eta^2 r^2) \right\}. \end{aligned} \quad (3.32 b)$$

Given the values of r_c and η for a particular simulation, these integrals can now be evaluated numerically and inserted into (2.21) from which $\varepsilon_L(k)$ and $\varepsilon_T(k)$ are then obtained.

In addition, these explicit results may also serve as a starting point for a more quantitative discussion of the Ewald sum itself. Thus it may be shown that in the limit $\eta \rightarrow \infty$ (r_c is usually fixed at $L/2$) the general expressions for the Ewald representation do indeed converge to the result for the idealized lattice sum with $\varepsilon_{\text{RF}} = \infty$, equation (3.27). This can be demonstrated by first removing the error function from the integrals by means of a partial integration, introducing $x = \eta r$ as a new variable of integration, expanding the Bessel functions into power series in $kr = kx/\eta$, and finally letting $\eta \rightarrow \infty$. In this way it is found that $2A(k) + B(k) \rightarrow -2$ (for $k \neq 0$) and $B(k) - A(k) \rightarrow 1$ (for all k). Since only a single term has been taken into account in the real space sum (3.30 *b-c*), taking the limit $\eta \rightarrow \infty$ corresponds to shifting the sum entirely into Fourier space.

On the other hand, we may ask under what conditions the formulas (3.27) for the idealized lattice sum might be used instead of (3.32), i.e. can we derive a criterion for the accuracy of the Ewald sum from (3.32)? To that end we first note that $B(k) - A(k)$ depends only on the real space sum. Since usually $r_c = L/2$, we may try to choose η such that $B(k) - A(k)$ is very close to unity, with a maximum deviation of 0.1 per cent, say. (Strictly speaking, this should hold for all k , but for the sake of simplicity we will only consider the case $k = 0$. Then our requirement is equivalent to the criterion of [8] that Q should be close to unity.) Having fixed η in this way we then note that, since $2A(k) + B(k)$ is to equal -2 , the sum of the two integrals on the right-hand side of (3.32 *a*) will be given by $3 \exp(-k^2/4\eta^2) - 2$, approximately. When combined with the first term on the right-hand side of (3.32 *a*) this of course adds up to -2 . However, due to the (spherical) cutoff in k -space the first term is only present for $0 < k \leq k_{\text{max}}$, so that for $k > k_{\text{max}}$ the value of $2A(k) + B(k)$ will be given by the sum of the integrals alone. In order for this sum to be sufficiently close to -2 , e.g. with a maximum deviation of 0.1 per cent again, we thus simply have to choose k_{max} such that $3 \exp(-k_{\text{max}}^2/4\eta^2) \lesssim 0.002$. (In this second part of the criterion the fact has been ignored that the integrals do not extend to infinity, but are truncated at r_c .)

As a specific example, we consider the pair of values characterizing the Ewald sum used in [15], i.e. $\eta L = 5.76$ and $k_{\text{max}} L/2\pi = 5$. We find $B(0) - A(0) = 0.99914$ and $3 \exp(-k_{\text{max}}^2/4\eta^2) = 0.00177$, indicating that in these simulations an excellent approximation has been used and that the formulas for the idealized lattice sum could be applied.

3.5. Characterization of boundary conditions

In the previous sections explicit fluctuation formulas have been derived for all boundary conditions presently used in simulations. Summarizing these results we are now in a position to discuss these simulation techniques from a common point of view, by concentrating on the k -dependent g -factors. In fact, it seems quite natural to classify boundary conditions according to whether the g -factors obtained in the simulations are identical to those of the infinite system or not. A summary of this kind is presented in tables 1 and 2, with table 1 giving the general overview and table 2 providing explicit results at zero wavevector, which will be used in our discussion of simulation results in § 4.

Table 1. Boundary conditions for which some of the finite-system g -factors are identical to those of the infinite system. (RF = reaction field, LS = lattice sum, EK = Ewald sum).

	RF($\epsilon_{\text{RF}} = \epsilon_0$)	RF($\epsilon_{\text{RF}} = \infty$)	LS($\epsilon_{\text{RF}} = \epsilon_0$)	LS($\epsilon_{\text{RF}} = \infty$), EK
$G^{\text{L}}(k) = g^{\text{L}}(k)$	No	No	Yes	Yes
$G^{\text{T}}(k) = g^{\text{T}}(k)$	No	Yes	Yes	Yes
$G^{\text{L}}(0) = g^{\text{L}}(0)$	No	No	No	No
$G^{\text{T}}(0) = g^{\text{T}}(0)$	No	Yes	No	Yes
$G(0) = g(0) = g_{\text{K}}$	Yes	No	Yes	No

According to table 1, if the simulation is to yield the same g -factors as the infinite system, then lattice summation would appear to be the method of choice. Here, all g -factors are reproduced correctly for $k \neq 0$, whereas the reaction field method—even if conducting boundary conditions ($\epsilon_{\text{RF}} = \infty$) are applied—will only give the correct $G^{\text{T}}(k)$. For $k = 0$, on the other hand, both simulation techniques are equivalent: By choosing appropriate values for ϵ_{RF} the infinite-system result can be obtained either for the total g -factor G_{K} ($\epsilon_{\text{RF}} = \epsilon_0$) or for its transverse component $G^{\text{T}}(0)$ ($\epsilon_{\text{RF}} = \infty$), but not for G_{K} and $G^{\text{T}}(0)$ simultaneously. By contrast, $G^{\text{L}}(0)$ cannot be made to coincide with $g^{\text{L}}(0)$ by any choice of boundary conditions. Interestingly, it is only the total g -factor g_{K} which may be obtained directly from reaction field geometry in the original sense ($\epsilon_{\text{RF}} = \epsilon_0$).

On the basis of these results one might be ready to conclude that simulations using a lattice sum are generally to be preferred over reaction field simulations, since they yield the same short-range structure as the infinite system—or at least the best approximation (we still have $G^{\text{L}}(0) \neq g^{\text{L}}(0)$). However, while the latter may still turn out to be true, we should not attach too much importance to a coincidence of the g -factors. After all, $g^{\text{L}}(k)$ and $g^{\text{T}}(k)$ are the (continuous) Fourier transforms of certain components of the pair correlation function of an *infinite*,

Table 2. Values of the finite-system g -factors at $k = 0$.
(SC = spherical cutoff, RF = reaction field, LS = lattice sum, EK = Ewald sum).

	G^{L}	G^{T}	$G = G^{\text{L}} + 2G^{\text{T}}$
Infinite system	$g^{\text{L}}(0) = \frac{1}{2\epsilon_0 + 1} g_{\text{K}}$	$g^{\text{T}}(0) = \frac{\epsilon_0}{2\epsilon_0 + 1} g_{\text{K}}$	$g(0) = g_{\text{K}}$
SC	$\frac{3\epsilon_0}{(2\epsilon_0 + 1)(\epsilon_0 + 2)} g_{\text{K}}$	$\frac{1}{3} g_{\text{K}}$	$\frac{9\epsilon_0}{(2\epsilon_0 + 1)(\epsilon_0 + 2)} g_{\text{K}}$
LS($\epsilon_{\text{RF}} = 1, k = 0$)			
RF($\epsilon_{\text{RF}} = \epsilon_0$)			
LS($\epsilon_{\text{RF}} = \epsilon_0, k = 0$)			
RF($\epsilon_{\text{RF}} = \infty$)			
LS($\epsilon_{\text{RF}} = \infty, k = 0$)	$\frac{\epsilon_0}{2\epsilon_0 + 1} g_{\text{K}}$	$\frac{\epsilon_0}{2\epsilon_0 + 1} g_{\text{K}}$	$\frac{3\epsilon_0}{2\epsilon_0 + 1} g_{\text{K}}$
EK($k = 0$)			
LS($k \rightarrow 0$)	$\frac{1}{2\epsilon_0 + 1} g_{\text{K}}$	$\frac{\epsilon_0}{2\epsilon_0 + 1} g_{\text{K}}$	g_{K}
EK($k \rightarrow 0$)			

isotropic sample, whereas $G^L(k)$ and $G^T(k)$ are (discrete) components of polarization fluctuations of a *finite, toroidal* system. Thus, even if we had $G^{L,T}(k) = g^{L,T}(k)$ for *all* allowed k -vectors, this would not necessarily imply that the $h_A(r)$ and $h_D(r)$ calculated from the simulation are identical to the correlation functions of an infinite system.

From a practical point of view it therefore seems to be much more important to emphasize that, using the formulas of §§ 2 and 3, the true dielectric constants $\varepsilon_L(k)$ and $\varepsilon_T(k)$, and hence the infinite-system g -factors $g^L(k)$ and $g^T(k)$, may now be calculated from *any* (reasonable) type of simulation. Since $g^L(k)$ and $g^T(k)$ are related to certain components of the pair correlation function, we have thus found a method by means of which information on the structure of the infinite system can be obtained, in an indirect way, from simulations of finite samples using more or less arbitrary boundary conditions. This will be discussed more fully in § 5.

4. DEPENDENCE UPON BOUNDARY CONDITIONS OF $G^L(k)$ AND $G^T(k)$

The logical next step in our study would be to show in a series of simulations that the dielectric constants $\varepsilon_L(k)$ and $\varepsilon_T(k)$ calculated by means of the formulas of §§ 2 and 3 are indeed independent of the boundary conditions employed in the simulation. However, from the extensive simulations reported in [11] we know already that this is true for the bulk dielectric constant $\varepsilon(k=0, \omega)$ where the effect of the boundary conditions is expected to be most pronounced. Therefore a different strategy will be adopted here: Starting from the g -factors obtained from a single simulation we will examine how $G^L(k)$ and $G^T(k)$ change with boundary conditions by calculating the g -factors that would have been obtained if different boundary conditions had been used. If our predictions should turn out to be in at least qualitative agreement with the behaviour observed in actual simulations, this might be taken as evidence that our formalism is correct.

In the literature simulation results for $\varepsilon(k)$ and $G(k)$ obtained with more than one type of boundary conditions and covering a wider range of k -values have only been reported by Pollock and Alder [2]. (In [3] and [4] only a few isolated k -vectors have been considered, and only the Ewald sum has been used.) Pollock and Alder have performed simulations for various Stockmayer systems, using Ewald as well as SC geometry. The particular system we will be referring to is that characterized by $\mu^* = 1.75$, $\rho^* = 0.8$, $T^* = 1.35$ and $\varepsilon_0 = 38$ (figures 9 and 10 of [2]). Unfortunately these authors do not give separate results for $G^L(k)$ and $G^T(k)$, but only for the total g -factor $G(k)$, so that our predictions can only be compared with the latter. Also, they do not calculate $\varepsilon_L(k)$ and $\varepsilon_T(k)$, but a *scalar* quantity $\varepsilon(k)$ (presumably obtained by inverting equation (2.13) of [2]).

We have therefore performed an additional simulation for the already extensively studied [7, 11] Stockmayer (= Lennard-Jones plus point dipole) system characterized by the reduced (with respect to the LJ ε and σ) parameters $\mu^{*2} = 3.0$, $\rho^* = 0.822$, and $T^* = 1.15$. With a bulk static dielectric constant of $\varepsilon_0 = 66$ this system is somewhat more strongly polar than that of Pollock and Alder ($\varepsilon_0 = 38$). The number of particles in the cubic basic cell was $N = 256$, and 40 000 time steps ($\Delta t^* = 0.0025$) have been performed in RF geometry with $\varepsilon_{\text{RF}} = \infty$, i.e. conducting boundary conditions. (For further details see [7, 11].)

$G^L(k)$ and $G^T(k)$ have only been calculated for (and averaged over equivalent) k -vectors parallel to the axes of the basic cell, i.e. for k an integer multiple of the

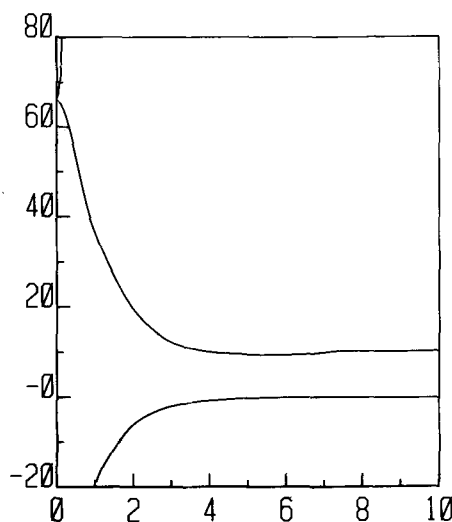


Figure 1. k -dependent longitudinal and transverse dielectric constant, $\varepsilon_L(k)$ and $\varepsilon_T(k)$, for a Stockmayer system with $\mu^{*2} = 3.0$ at $\rho^* = 0.822$ and $T^* = 1.15$. Continuous curve: $\varepsilon_T(k)$; curve with singularity: $\varepsilon_L(k)$. k in reduced units, $k^* = k\sigma$.

smallest non-zero k -vector $k_{\min}^* = 2\pi/L^* = 0.927$. Intermediate values have been obtained by cubic interpolation between these grid points. Wavevectors with $0 < k < k_{\min}$ are not directly accessible in the simulation (unless the size of the system were increased), but for $k > k_{\min}$ the accuracy of the results could be improved by using all k -vectors that are compatible with the basic cell.

Using the finite-system g -factors $G^L(k)$ and $G^T(k)$ from the simulation and $A(k)$ and $B(k)$ for this particular geometry, i.e. equation (3.17) with $\varepsilon_{\text{RF}} = \infty$, the dielectric constants $\varepsilon_L(k)$ and $\varepsilon_T(k)$ can now be evaluated by means of (2.21). The result is shown in figure 1. We first note that (essentially by construction) $\varepsilon_L(k) \rightarrow \varepsilon_0$ and $\varepsilon_T(k) \rightarrow \varepsilon_0$ for $k \rightarrow 0$. The transverse dielectric constant then drops very quickly—at k -values corresponding to a wavelength of $\sim 2\sigma$ —to its Onsager limit of $\varepsilon_T(k \rightarrow \infty) = 9.98$ (cf. (3.7)), around which there are hardly any oscillations. The longitudinal dielectric constant, on the other hand, according to (3.5) has to obey either $\varepsilon_L(k) \geq 1$ or $\varepsilon_L(k) \leq 0$ and hence cannot cross the strip $0 < \varepsilon < 1$. Therefore, in order to reach the (negative) Onsager limit of $\varepsilon_L(k \rightarrow \infty) = -0.125$, it has to pass through a singularity. (A similar phenomenon is also found in the dielectric constant of ionic liquids [16].) In our case the singularity occurs at some $k < k_{\min}$, where only interpolated values are available, and so cannot be analysed in more detail. The oscillations around the asymptotic Onsager value are comparatively larger than in the case of the transverse dielectric constant, but cannot be discerned on the scale of the figure.

The g -factors $g^L(k)$, $g^T(k)$, and $g(k)$ for the *infinite* system, obtained with the aid of (3.3) from the dielectric constant of figure 1, are displayed in figure 2. The total g -factor $g(k)$ is similar to that reported by Pollock and Alder (figure 9 of [2]). Starting from the Kirkwood g -factor $g(0) = g_K = 4.9$ at $k = 0$, it quickly drops to the Onsager limit $g(k \rightarrow \infty) = 1$ and then oscillates around the latter. Of greater interest is its decomposition into longitudinal and transverse components, which seems not to have been given before in the literature. The structure of $g^T(k)$ is

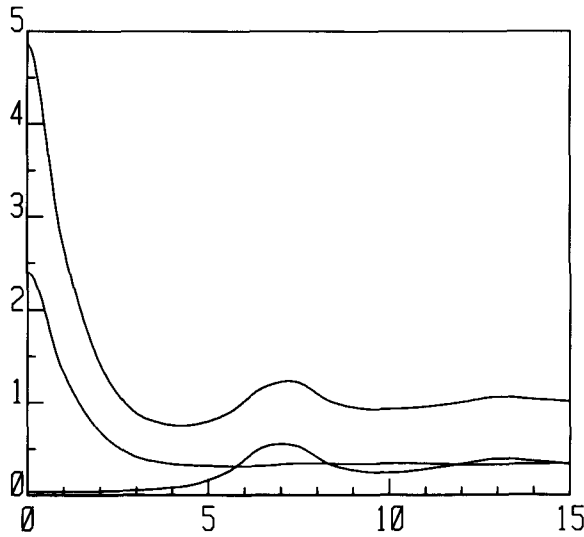


Figure 2. k -dependent infinite-system g -factors $g^L(k)$, $g^T(k)$, and $g(k)$ for the Stockmayer system of figure 1. Left side top to bottom: $g(k)$, $g^T(k)$, and $g^L(k)$, interpolated between the values calculated from the dielectric constant for k an integer multiple of $k_{\min}^* = 2\pi\sigma/L = 0.927$. The curves of this figure have also been used to interpolate the dielectric constant of figure 1 and the g -factors of figures 3–6.

relatively simple: Starting at roughly one half of the Kirkwood g -factor (cf. table 2, for $\varepsilon_0 \gg 1$), it also drops quickly and monotonically to the Onsager limit $1/3$, remaining virtually constant for $k^* \gtrsim 3$ –4. Note that, due to our choice of boundary conditions (RF geometry with $\varepsilon_{\text{RF}} = \infty$, cf. (3.21)), the $g^T(k)$ shown in figure 2 for all k is identical to the finite-system g -factor $G^T(k)$ as obtained from the simulation. The longitudinal g -factor $g^L(k)$, on the other hand, rises rather slowly from a very small value at $k = 0$ (cf. table 2) and then exhibits a series of only weakly damped oscillations before eventually reaching the limit $1/3$. Thus at small k the shape of the total g -factor is essentially determined by $g^T(k)$, but for large k the influence of $g^L(k)$ is dominant.

In figures 3–5 we present *finite*-system g -factors $G^L(k)$, $G^T(k)$, and $G(k)$ for various toroidal geometries, in order to demonstrate the influence of the boundary conditions. These g -factors have been obtained from the dielectric constant of figure 1, by using equation (2.21) with the $A(k)$ and $B(k)$ appropriate for the geometry in question. Only RF and SC geometry have been considered, since for $k \neq 0$ the g -factors from lattice sum (Ewald) simulations are identical to the infinite-system g -factors anyway. All curves are for a system of $N = 256$ particles with $r_c = L/2$ and, strictly speaking, only the results represented by symbols (circles, triangles etc.) should have been included, as these correspond to the discrete set of k -vectors accessible in the simulation. However, for clarity these points have been joined by continuous curves (dashed) which were obtained from the interpolated infinite-system g -factors $g^L(k)$ and $g^T(k)$ by simply using (3.3) and (2.21) for arbitrary k †.

† These curves would correspond to the g -factors of an *infinite* system with a *finite*-ranged (modified) interaction truncated at $r = r_c$.

In figure 3 the total g -factor $G(k)$ for a system of 256 particles in SC and RF geometry with $\epsilon_{\text{RF}} = \epsilon_0$ and $\epsilon_{\text{RF}} = \infty$ (symbols and dashed lines) is compared with the total g -factor $g(k)$ for an infinite sample (solid line). Between $k = 0$ and $k = k_{\text{min}}$ the behaviour of the curves for $\epsilon_{\text{RF}} = \epsilon_0$ and $\epsilon_{\text{RF}} = \infty$ should not be taken too seriously, since it mainly reflects our way of interpolating the raw data. For

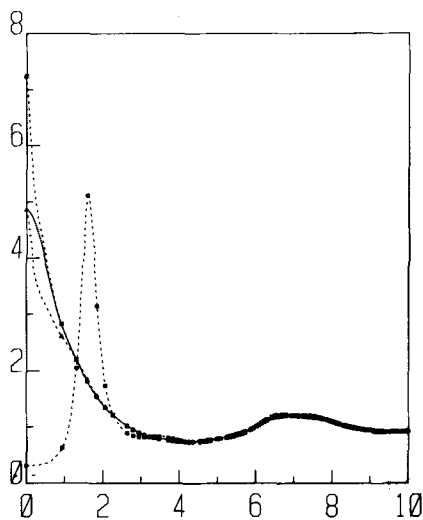


Figure 3. Dependence upon boundary conditions of the total g -factor $G(k)$ for the Stockmayer system of figure 1. Solid line: $g(k)$ for the infinite system; symbols and dashed lines: hypothetical $G(k)$ for a system of 256 particles in SC geometry (circles) and RF geometry with $\epsilon_{\text{RF}} = \epsilon_0$ (triangles) and $\epsilon_{\text{RF}} = \infty$ (squares). Only the values marked by symbols, corresponding to the k -vectors accessible in the simulation, can be observed. Results for LS and EK geometry have not been included since there $G^{\text{L},\text{T}}(k) = g^{\text{L},\text{T}}(k)$, except at $k = 0$.

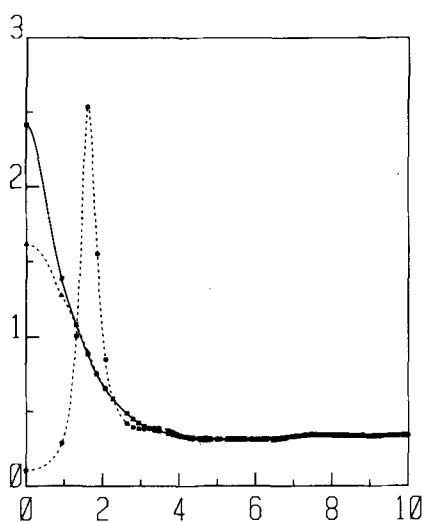


Figure 4. Dependence upon boundary conditions of the transverse g -factor $G^{\text{T}}(k)$ for the Stockmayer system of figure 1. Lines and symbols as in figure 3. Note that for RF geometry with $\epsilon_{\text{RF}} = \infty$ $G^{\text{T}}(k) = g^{\text{T}}(k)$ for all k .

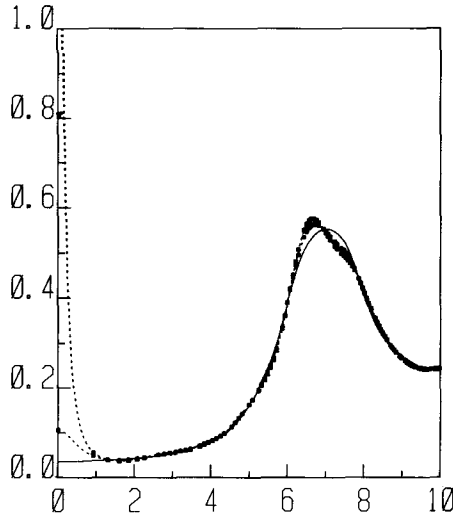


Figure 5. Dependence upon boundary conditions of the longitudinal g -factor $G^L(k)$ for the Stockmayer system of figure 1. Lines and symbols as in figure 3.

$k \gtrsim \sqrt{2k_{\min}}$ the g -factors obtained in RF geometry are already in good agreement with $g(k)$ for the infinite system. At the first two k -vectors, however, we can only have either $G(0) = g(0) = g_K$ (for $\varepsilon_{\text{RF}} = \varepsilon_0$) or $G(k_{\min}) \approx g(k_{\min})$ (for $\varepsilon_{\text{RF}} = \infty$, in which case $G(0) \simeq (3/2)g_K$), but not both. The g -factor for SC geometry, on the other hand, presents a completely different picture. Apart from being much smaller at $k = 0$ ($G(0) \simeq (9/2\varepsilon_0)g_K$, according to table 2), it also exhibits a pronounced maximum at $k = \sqrt{3k_{\min}}$ —clearly an artifact of the boundary conditions! Qualitatively, this g -factor is very similar to that reported by Pollock and Alder for an actual simulation in SC geometry (figure 9 of [2]), but since their model system is slightly different from ours (differing also by their use of a tapering function, which could easily be included into our theory), no quantitative comparison can be made. Nevertheless we feel encouraged to believe that the influence of the boundary conditions on k -dependent dielectric properties is correctly described by our theory.

In figures 4 and 5 $G(k)$ has been decomposed into transverse and longitudinal components. From figure 4 it is obvious that the spurious maximum in $G(k)$ for SC geometry is due to the transverse component $G^T(k)$. In RF geometry with conducting boundary conditions ($\varepsilon_{\text{RF}} = \infty$) the transverse component for the infinite system, $g^T(k)$, can be obtained directly from the simulation, $G^T(k) = g^T(k)$ (cf. (3.21)). However, if the ‘correct’ reaction field ($\varepsilon_{\text{RF}} = \varepsilon_0$) is used, small deviations from $g^T(k)$ will result; in particular we obtain $G^T(0) \simeq g_K/3$ instead of $g^T(0) \simeq g_K/2$ at $k = 0$. Turning now to the longitudinal component (figure 5) we note that for $k > k_{\min}$ the differences between the various geometries are small. In SC geometry $G^L(0) \simeq (3/2\varepsilon_0)g_K$ lies somewhat above infinite-system result $g^L(0) \simeq (1/2\varepsilon_0)g_K$. Upon including a reaction field the values are considerably larger, i.e. (not visible in figure 5) $G^L(0) \simeq g_K/3$ for $\varepsilon_{\text{RF}} = \varepsilon_0$ and $G^L(0) \simeq g_K/2$ for $\varepsilon_{\text{RF}} = \infty$. Additional slight deviations of the finite-system results from the longitudinal g -factor for the infinite system are also noticeable around the main peak of $G^L(k)$ at $k^* \sim 7$.

5. THE STRUCTURE OF THE INFINITE SYSTEM

In the previous sections we have shown how the correct dielectric constants $\varepsilon_L(k)$ and $\varepsilon_T(k)$ as well as the g -factors $g^L(k)$, $g^T(k)$, and $g(k)$ for an infinite system may be calculated from a simulation of a finite sample. Also, we have shown how the g -factors $G^L(k)$, $G^T(k)$, and $G(k)$ for finite systems with various boundary conditions can be translated into one another. In this final section we will now outline a method by means of which the structure of the infinite system itself may be obtained from a simulation with arbitrary boundary conditions.

The structure of a dielectric is probably best characterized by the R -dependent generalization of the Kirkwood g -factor

$$G_K(R) = 1 + \frac{4\pi\rho}{3} \int_0^R dr r^2 h_\Delta(r). \quad (5.1)$$

$G_K(R)$ is the total dipole moment (in units of the molecular dipole moment μ) of a sphere of radius R around a reference particle. In an infinite system $G_K(R)$ tends to the Kirkwood factor g_K as $R \rightarrow \infty$ and thus describes how the individual shells of neighbours around the reference particle contribute to g_K .

According to (3.11)–(3.13) $g(k)$ essentially is the Fourier transform of $h_\Delta(r)$ for an infinite system. On the other hand, we know from §§ 2 and 3 that $g(k)$ for an infinite system may be obtained from a simulation by first calculating $\varepsilon_L(k)$ and $\varepsilon_T(k)$ from the finite-system g -factors $G^L(k)$ and $G^T(k)$ (equation (2.21)) and then using (3.3) to find $g^L(k)$ and $g^T(k)$. By simply performing an inverse Fourier transform on $g(k) = g^L(k) + 2g^T(k)$ (to get $h_\Delta(r)$) and inserting into (5.1) we should thus be able to calculate the true R -dependent g -factor for an infinite system, and the result should be independent of the boundary conditions used in the simulation.

As a first test we have therefore tried to calculate the structure of the infinite system from the reaction field simulation with conducting boundary conditions of § 4. The result is shown in figure 6, where our hypothetical $G_K(R)$ for the infinite system (circles) is compared with the finite-system g -factors from previous reaction field simulations with $N = 256$ and $\varepsilon_{\text{RF}} = 50$ [7], and $N = 512$ and $\varepsilon_{\text{RF}} = 70$ [11], respectively (i.e. $\varepsilon_{\text{RF}} \simeq \varepsilon_0$ in both cases), and seems to be very encouraging indeed: In order to calculate the points of figure 6 only 18 values ($k = 0, k_{\min}, \dots, 17k_{\min}$; with cubic interpolation) of $g(k)$ have been used for the inverse Fourier transform. Including higher k -values does not change the results qualitatively. Although, in a more accurate calculation, a finer grid should be used for $k > k_{\min}$ as well as a more sophisticated interpolation for $0 < k < k_{\min}$, using just the first few points of $g(k)$ seems to be sufficient to reproduce the R -dependent g -factor not only qualitatively but also quantitatively. In particular, the contributions from the first two shells of neighbours are predicted correctly, as is the asymptotic approach to g_K . The fact that the R -dependent g -factor for the infinite system for small R always lies below that of a finite system with $\varepsilon_{\text{RF}} \simeq \varepsilon_0$ seems to be correct too. In a finite system the asymptotic value $G_K \simeq g_K$ has to be reached for $R \rightarrow \sqrt{3}L/2$, i.e. over a distance which not only depends on N but may also be smaller than the correlation length in an infinite system. Thus $G_K(R)$ for a finite system may be systematically too high in this range. In our case the correlation length of the infinite system does seem to be greater than (or at least comparable to) the dimensions of the sample used in the simulation ($\sqrt{3}L/2 \sim 6\sigma$). It should

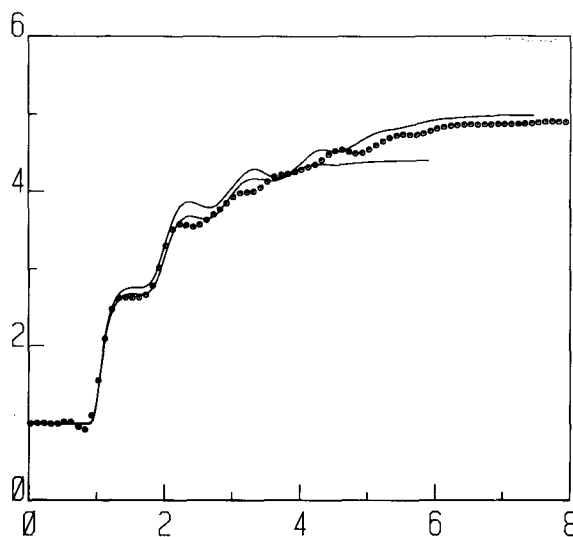


Figure 6. R -dependent Kirkwood g -factor $G_K(R)$ for the Stockmayer system of figure 1. Circles: hypothetical $G_K(R)$ for an infinite system, obtained from the inverse Fourier transform of the k -dependent g -factors of figure 2. Solid lines: finite-system $G_K(R)$ from RF simulations with $\varepsilon_{\text{RF}} = 70$, $N = 512$ [11] (lower curve for $R^* \lesssim 4$, upper curve for $R^* \gtrsim 4$) and $\varepsilon_{\text{RF}} = 50$, $N = 256$ [7] (upper curve for $R^* \lesssim 4$, lower curve for $R^* \gtrsim 4$). R in reduced units, $R^* = R/\sigma$.

also be pointed out that the R -dependent g -factor $G_K(R)$ obtained directly from a simulation with conducting boundary conditions for large R would lie well above (by a factor of $\sim 3/2$) the curves of figure 6 (cf. figure 2 of [7] and figure 6 of [11]). However, by performing the 'deconvolution' in k -space this effect of the boundary conditions may be eliminated.

We have thus shown for the first time how, starting from a computer simulation with one of the typical boundary conditions, the correct structure of the infinite system can be obtained. Of course, the theory presented in this article and in particular the deconvolution discussed above both rest on the assumption that the sample, although finite, is sufficiently large so that only the effect of the boundary conditions on the structure induced by the *longest*-ranged (i.e. dipolar) interactions need to be taken into account. Thus the method cannot be expected to explain the changes (possibly also due to non-dipolar orientation-dependent interactions) in the structure of samples of extremely small size. Yet, even if our indirect method should not turn out to be practical in general and for arbitrary model systems, the mere possibility of eliminating (at least theoretically) the influence of the boundary conditions also in r -space illustrates the progress we have made in understanding the behaviour of polar systems in computer simulations.

APPENDIX

Connection of k -dependent g -factors with the pair correlation function

In order to make this article self-contained, we include a simple derivation of equations (3.11–12) of the main text. These results have also been given in [1] and [2].

The k -dependent g -factors of the infinite system may be obtained from those of a finite system, equation (2.22), by letting $N, V \rightarrow \infty$

$$g^L(k) = \lim_{N, V \rightarrow \infty} \left\langle \sum_{i,j=1}^N (\hat{\mathbf{k}} \cdot \boldsymbol{\mu}_i)(\hat{\mathbf{k}} \cdot \boldsymbol{\mu}_j) \exp(-i\mathbf{k} \cdot \mathbf{r}_{ij}) \right\rangle / N\mu^2, \quad (\text{A } 1 \text{ } a)$$

$$g^T(k) = \lim_{N, V \rightarrow \infty} \left\langle \sum_{i,j=1}^N [\boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j - (\hat{\mathbf{k}} \cdot \boldsymbol{\mu}_i)(\hat{\mathbf{k}} \cdot \boldsymbol{\mu}_j)] \exp(-i\mathbf{k} \cdot \mathbf{r}_{ij}) \right\rangle / 2N\mu^2. \quad (\text{A } 1 \text{ } b)$$

(Taking the limit $N, V \rightarrow \infty$ before letting $k \rightarrow 0$ ensures that the results will also be correct for $k = 0$.) From (A 1) we can form the linear combinations

$$g(k) = g^L(k) + 2g^T(k) = \lim_{N, V \rightarrow \infty} \left\langle \sum_{i,j} \boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j \exp(-i\mathbf{k} \cdot \mathbf{r}_{ij}) \right\rangle / N\mu^2, \quad (\text{A } 2 \text{ } a)$$

$$2[g^L(k) - g^T(k)] = \lim_{N, V \rightarrow \infty} \left\langle \sum_{i,j} \boldsymbol{\mu}_i \cdot (3\hat{\mathbf{k}}\hat{\mathbf{k}} - \mathbf{I}) \cdot \boldsymbol{\mu}_j \exp(-i\mathbf{k} \cdot \mathbf{r}_{ij}) \right\rangle / N\mu^2. \quad (\text{A } 2 \text{ } b)$$

In an infinite, isotropic system $g^L(k)$ and $g^T(k)$ will depend only on the magnitude of $\mathbf{k}$. Thus we may insert an additional average over all orientations $\Omega_{\mathbf{k}}$ of $\mathbf{k}$ into (A 2) *before* performing the ensemble average and the limit $N, V \rightarrow \infty$. Using the Rayleigh expansion for $\exp(-i\mathbf{k} \cdot \mathbf{r})$ again (cf. § 2) we have

$$\frac{1}{4\pi} \int d\Omega_{\mathbf{k}} \exp(-i\mathbf{k} \cdot \mathbf{r}) = j_0(kr), \quad (\text{A } 3 \text{ } a)$$

$$\frac{1}{4\pi} \int d\Omega_{\mathbf{k}} \exp(-i\mathbf{k} \cdot \mathbf{r})(3\hat{\mathbf{k}}\hat{\mathbf{k}} - \mathbf{I}) = -(3\hat{\mathbf{r}}\hat{\mathbf{r}} - \mathbf{I})j_2(kr) \quad (\text{A } 3 \text{ } b)$$

so that

$$g^L(k) + 2g^T(k) = \lim_{N, V \rightarrow \infty} \left\langle \sum_{i,j} \boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j j_0(kr_{ij}) \right\rangle / N\mu^2, \quad (\text{A } 4 \text{ } a)$$

$$\begin{aligned} & 2[g^L(k) - g^T(k)] \\ &= - \lim_{N, V \rightarrow \infty} \left\langle \sum_{i,j} [3(\hat{\mathbf{r}}_{ij} \cdot \boldsymbol{\mu}_i)(\hat{\mathbf{r}}_{ij} \cdot \boldsymbol{\mu}_j) - \boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j] j_2(kr_{ij}) \right\rangle / N\mu^2. \end{aligned} \quad (\text{A } 4 \text{ } b)$$

These expressions now contain averages of the angular functions $\Delta(\Omega_i, \Omega_j) = (\boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j)/\mu^2$ and $D(\Omega_i, \Omega_j) = [3(\hat{\mathbf{r}}_{ij} \cdot \boldsymbol{\mu}_i)(\hat{\mathbf{r}}_{ij} \cdot \boldsymbol{\mu}_j) - \boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j]/\mu^2$ which define the projections [17]

$$h_A(r) = \frac{3}{\Omega^2} \int d\Omega_1 d\Omega_2 g(r, \Omega_1, \Omega_2) \Delta(\Omega_1, \Omega_2), \quad (\text{A } 5 \text{ } a)$$

$$h_D(r) = \frac{3/2}{\Omega^2} \int d\Omega_1 d\Omega_2 g(r, \Omega_1, \Omega_2) D(\Omega_1, \Omega_2) \quad (\text{A } 5 \text{ } b)$$

of the angle-dependent pair correlation function $g(r, \Omega_1, \Omega_2)$. Ω_i and Ω_j are the relative (with respect to $\mathbf{r}_{ij}$) orientations of particles i and j , and $\Omega = \int d\Omega$. In the

limit $N, V \rightarrow \infty$ (A 4) may thus be written as

$$g(k) = g^L(k) + 2g^T(k) = 1 + \frac{4\pi\rho}{3} \int_0^\infty dr r^2 h_A(r) j_0(kr), \quad (\text{A } 6 \text{ } a)$$

$$g^L(k) - g^T(k) = -\frac{4\pi\rho}{3} \int_0^\infty dr r^2 h_D(r) j_2(kr). \quad (\text{A } 6 \text{ } b)$$

(Note that the self-terms of (A 4 b) do not contribute to (A 6 b) since $j_2(kr_{ij}) = 0$ for $i = j$.) Equations (3.11)–(3.14) now follow from (A 6).

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